

UMC 0.11um RFCMOS AI Advanced Enhancement 1.5fF/um² MIM Capacitors RF SPICE Model Document

**Version: 0.2 Phase 2
2013/8/19**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
v0.2p1	7/22/2010	Bing Chang Wu	Original release
v0.2p2	8/19/2013	Bing Chang Wu	Typo update

1.2 General Information

1.5fF/um² Planar MIM Capacitor RF Sub-circuit Model

Model File Name:

Spectre: L110AE_MIMCAPS_1P5F_RF_V022.lib.scs,L110AE_MIMCAPS_1P5F_RF_V022.mdl.scs

HSPICE:L110AE_MIMCAPS_1P5F_RF_V022.lib,L110AE_MIMCAPS_1P5F_RF_V022.mdl

Eldo:L110AE_MIMCAPS_1P5F_RF_V022.lib.eldo,L110AE_MIMCAPS_1P5F_RF_V022.mdl.eldo

Sub-circuit Model Name and Node Definition:

mimcaps_1p5f_psub_rfkf (top bottom psub) - for 3-terminal RF model with underneath native p-type substrate.

mimcaps_1p5f_nwell(top bottom nwell psub) - for 4-terminal RF model with underneath n-well.

mimcaps_1p5f_mm (top bottom) - for simplified 2-terminal model which represents the capacitance without parasitic component.

Corner Section:

ff: minimum capacitance value

tt: typical capacitance value

ss: maximum capacitance value

Input Parameters:

l : drawn length in m for a unit

w : drawn width in m for a unit

mo: metal option (1~26, default is 21)

1.3 Device Structure

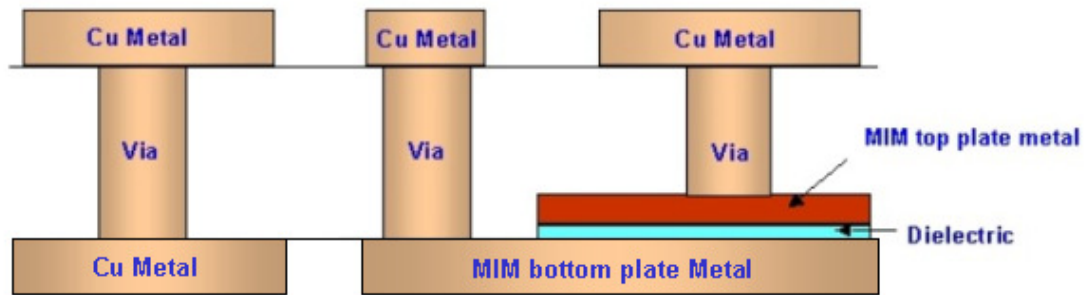


Figure 1-1 Planar Metal-Insulator-Metal (MIM) Capacitor cross-section

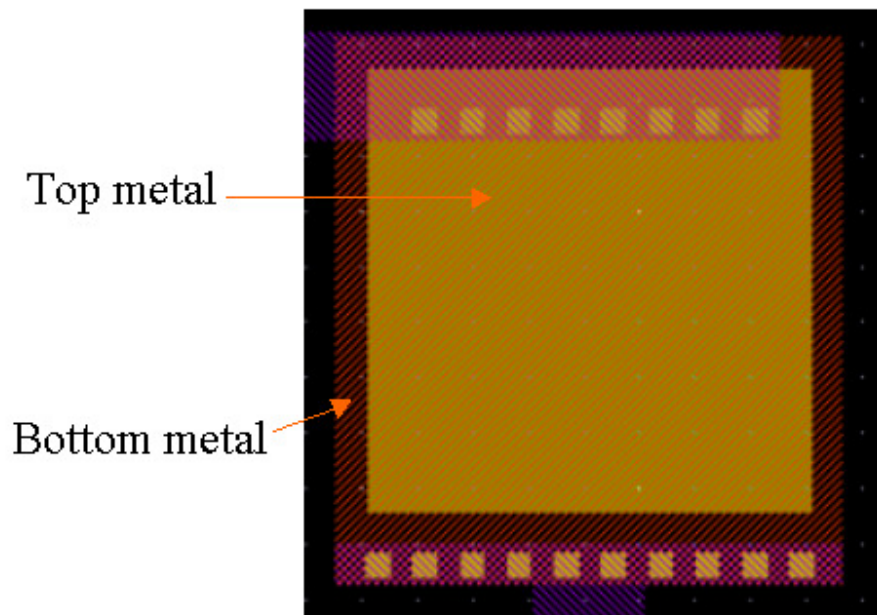


Figure 1-2 Layout Example

1.4 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

1.4.1 For Cadence Spectre

- Copy L110AE_MIMCAPS_1P5F_RF_V022.lib.scs and L110AE_MIMCAPS_1P5F_RF_V022.mdl.scs into your simulation directory
- '- Add the following statement in the input netlist to the simulator:
include "/ L110AE_MIMCAPS_1P5F_RF_V022.lib.scs"
section=CornerSection, where the CornerSection could be tt, ff or ss
- '- Follow the following example to call the model:
x1 (top bottom psub) mimcaps_1p5f_psub_rfkf l=10e-6 w=10e-6 mo=21

1.4.2 For Synopsys Hspice

- Copy L110AE_MIMCAPS_1P5F_RF_V022.lib and L110AE_MIMCAPS_1P5F_RF_V022.mdl into your simulation directory.
- '- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_MIMCAPS_1P5F_RF_V022.lib" CornerSection, where CornerSection could be tt, ff, ss.
- '- Follow the example below to call the model x1 top bottom psub
mimcaps_1p5f_psub_rfkf l=10e-6 w=10e-6 mo=21

1.4.3 For Mentor Graphics Eldo

- Copy L110AE_MIMCAPS_1P5F_RF_V022.lib.eldo and L110AE_MIMCAPS_1P5F_RF_V022.mdl.eldo into your simulation directory.
- '- Add the following statement in the input netlist to the simulator
.lib "/ L110AE_MIMCAPS_1P5F_RF_V022.lib.eldo " CornerSection, where CornerSection could be tt, ff, ss.
- '- Follow the example below to call the model x1 top bottom psub
mimcaps_1p5f_psub_rfkf l=10e-6 w=10e-6 mo=21

1.4.4 Model Usage Guideline

The model is generated for 1.5fF/um² Metal-Insulator-Metal (MIM) Capacitor, which will be manufactured by 110nm Mixed-Mode/RF Process.

Variable parameter are : w (drawn width of MIM Cap), l (drawn length of MIM Cap), and mo (metal option)

The following scale range (before 90% shrinkage) are recommended for the RF model.

w : 3um ~ 50um

l : 3um ~ 50um

mo : 1 ~ 26

w/l ratio is restricted from 0.5 to 2.0

Metal Options :

mo	Metal Options	1x metal	2x metal	Top metal (8k, 12k, 20k or 40k)	MIM Cap layer	MIM Cap layer
26	1P8M	M2-M6	M7	M8	M6	M6-M7
21	1P7M	M2-M5	M6	M7	M5	M5-M6
15	1P7M	M2-M6	M7	-	M6	M6-M7
12	1P6M	M2-M4	M5	M6	M4	M4-M5
9	1P6M	M2-M5	M6	-	M5	M5-M6
6	1P5M	M2-M3	M4	M5	M3	M3-M4
3	1P5M	M2-M4	M5	-	M4	M4-M5
1	1P4M	M2-M3	M4	-	M3	M3-M4

The MM model (mimcaps_1p5f_mm) predicts the capacitance without parasitic component, thus it is recommended for the application at low frequency.

The RF model (mimcaps_1p5f_psub_rfkf) is the 3-terminal RF model with the underneath of native p-type substrate.

The RF model (mimcaps_1p5f_nwell_rfkf) is the 4-terminal RF model with the underneath of n-well.

The valid frequency range for RF model is up to 20 GHz.

The valid temperature is following: -55 °C ~ +125 °C.

The square sharp is recommended.

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

2 RFMIM Capacitors

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafers information

BEOL option	1P7M
Mask ID	bsw0fa
Lot ID	D4WC4
Wafer ID	3

2.2 Equivalent Circuit

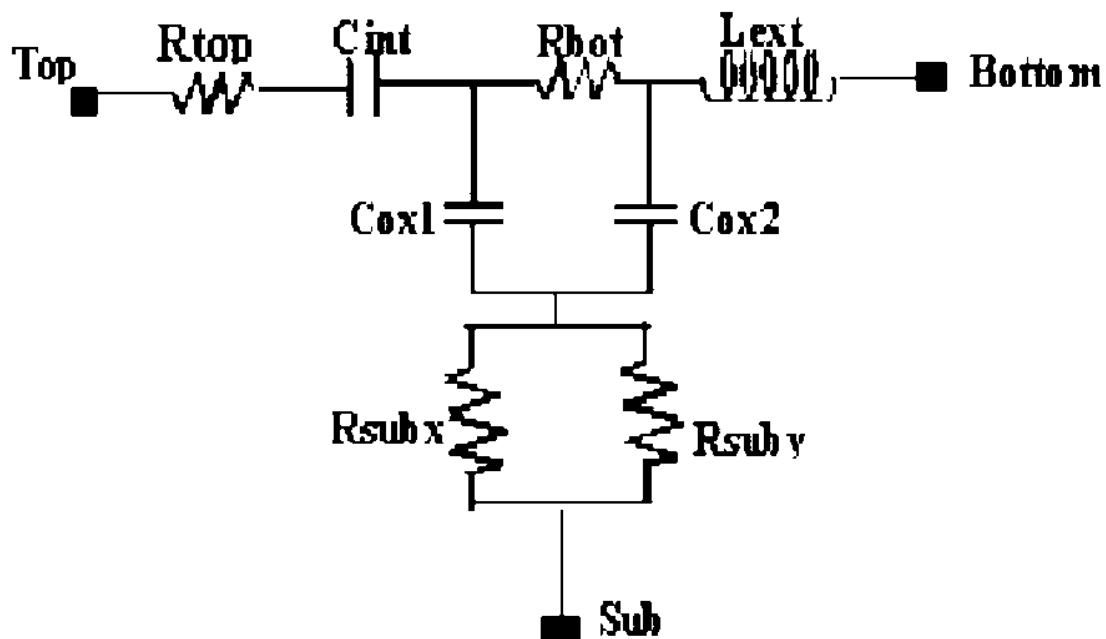


Figure 2-1 Equivalent circuit for mimcaps_1p5f_psub_rfkf

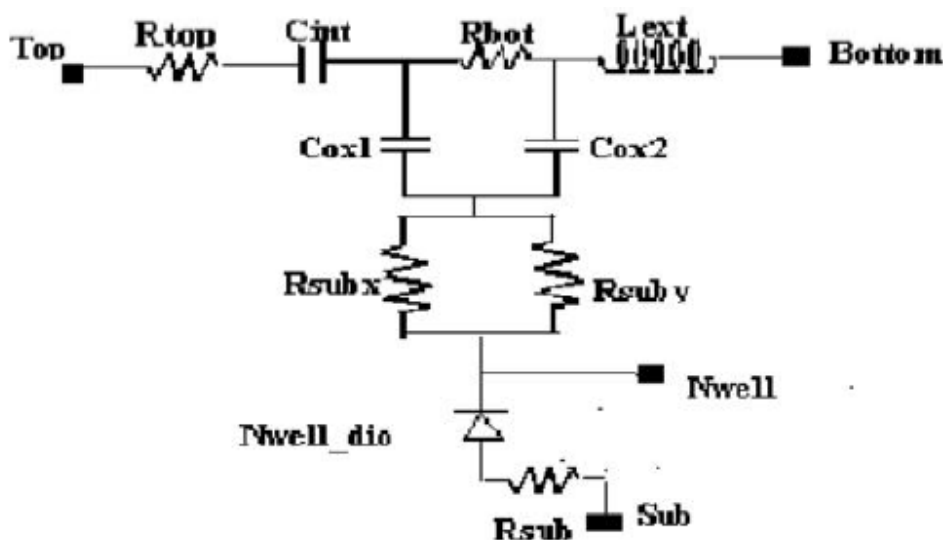


Figure 2-2 Equivalent circuit for mimcaps_1p5f_nwell_rfkf

2.3 Model Centering and Corner Model

Typical and Corner models are generated based on recent metal-insulator-metal capacitor characterization report.

Table 2-2 Parameter Spec.

Electrical Parameters	Typical value
Area component of capacitance (C_Area)	1.5266 fF/um ²
Perimeter component of capacitance (C_Peri)	0.0496 fF/um
Temperature coefficient (Tc1)	16.7 ppm/°C
Temperature coefficient (Tc2)	-0.007 ppm/°C ²
Voltage coefficient (Vc1)*	-9.05 ppm/V
Voltage coefficient (Vc2)	7.53 ppm/V ²

*Voltage drop across capacitor is defined as (Vtop_Vbottom)

Table 2-3 Process corner range

Parameter	Minimum	Maximum
C_area	Nominal x 0.85	Nominal x 1.15
C_peri	Nominal x 0.7	Nominal x 1.3
Rtop	Nominal x 1.3	Nominal x 0.7
Rbot	Nominal x 1.3	Nominal x 0.7
Cox1	Nominal x 1.15	Nominal x 0.85
Cox2	Nominal x 1.15	Nominal x0.85

2.4 Model Verification

All measurement results are denoted by .m and all simulation results are denoted by .s
Port1 and Port2 are connected to bottom plate and top plate respectively in S-parameter measurement and simulation.

Critical parameter definition:

o Main capacitance: $C_{12} = \frac{1}{\omega \cdot \text{Im}\left(\frac{1}{Y_{12}}\right)}$, where ω is the angular frequency.

o Bottom plate to substrate capacitance: $C_{11} = \frac{-1}{\omega \cdot \text{Im}\left(\frac{1}{Y_{11} + Y_{12}}\right)}$, where ω is the angular frequency.

QUALITY FACTOR: $Q = \frac{-\text{Im}\left(\frac{1}{Y_{11}}\right)}{\text{Re}\left(\frac{1}{Y_{11}}\right)}$

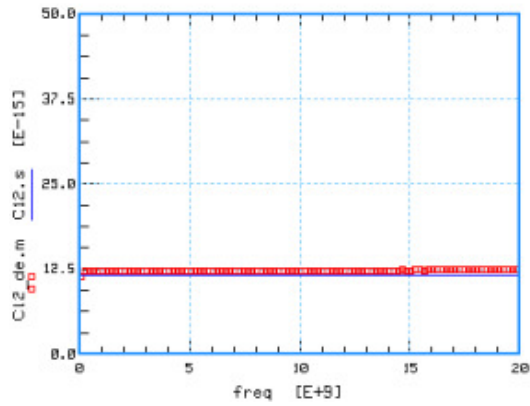
Table 2-4 Model verification in main capacitance and quality factor at f=5GHz for 1P7M2T

	drawn			measured at f=5GHz		extracted at f=5GHz		centered at f=5GHz	
Device	L(um)	W(um)	Substrate	C12	Q	C12	Q	C12	Q
MMC2	3	3	P-sub	1.22E-14	123	1.16E-14	121	1.14E-14	123
MMC3	5	5	P-sub	3.21E-14	103	3.18E-14	106	3.12E-14	108
MMC5	10	10	P-sub	1.26E-13	91.5	1.26E-13	92.3	1.23E-13	94
MMC5	30	30	P-sub	1.14E-12	69.4	1.14E-12	70.6	1.12E-12	71.9
MMC6	50	50	P-sub	3.37E-12	59.2	3.42E-12	54.9	3.35E-12	55.9
MMC2nw	3	3	N-well	1.20E-14	122	1.16E-14	122.2	1.14E-14	124
MMC3nw	5	5	N-well	3.23E-14	102	3.18E-14	107.4	3.12E-14	109
MMC3nw	10	10	N-well	1.28E-13	91	1.26E-13	93.3	1.23E-13	94.9
MMC5nw	30	30	N-well	1.12E-12	73.5	1.14E-12	71.4	1.12E-12	72.7
MMC6nw	50	50	N-well	3.39E-12	58.9	3.42E-12	55.6	3.34E-12	56.7

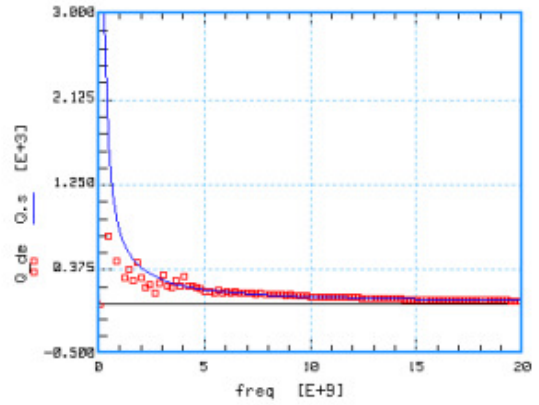
Table 2-5 Model-predicted main (C12), bottom plate (C11) capacitances and quality factor in different metal options (W/L=50um/50um, frequency=5GHz)

mo	C12.s(F)	C11.s(F)
26	3.35E-12	2.65E-14
21	3.35E-12	3.25E-14
15	3.35E-12	2.65E-14
12	3.35E-12	4.21E-14
9	3.35E-12	3.25E-14
6	3.35E-12	5.96E-14
3	3.35E-12	4.21E-14
1	3.35E-12	5.96E-14

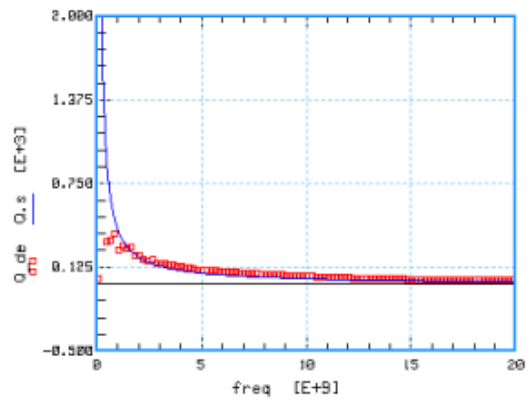
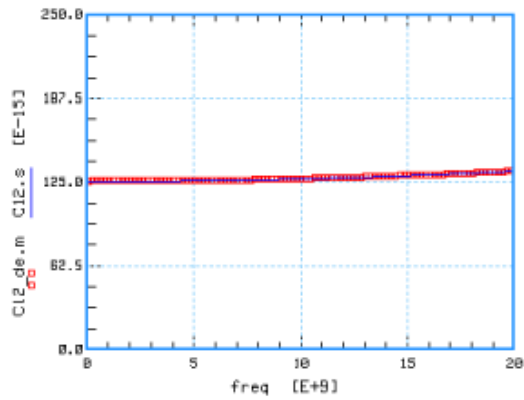
Main Capacitance



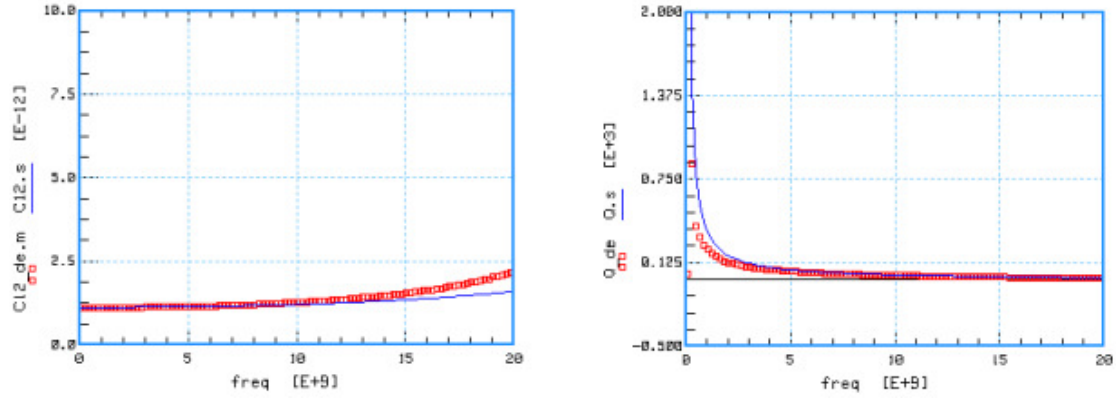
Quality factor



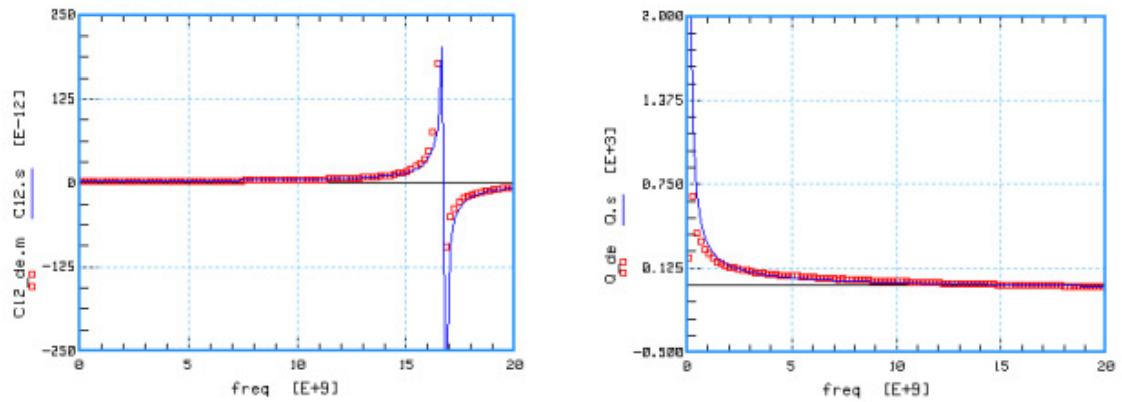
MMC2 : 3um x 3um



MMC4 : 10um x 10um



MMC5 : 30um x 30um



MMC6 : 50um x 50um

Figure 2-3 Model-predicted Q-factor and main capacitance at T=25 °C, frequency=500MHz~20GHz, metal option is 1P7M(mo=21)

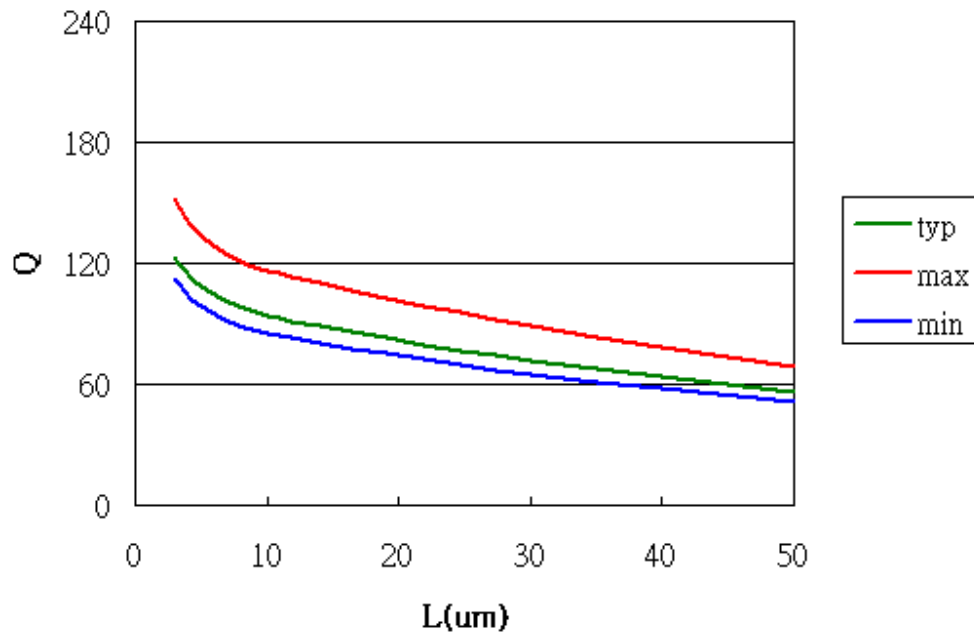


Figure 2-4 Model-predicted Q factor for 3-corner cases at T=25 °C, frequency=5GHz, W/L ratio=1, width range is 3um~50um, metal option is 1P7M(mo=21)

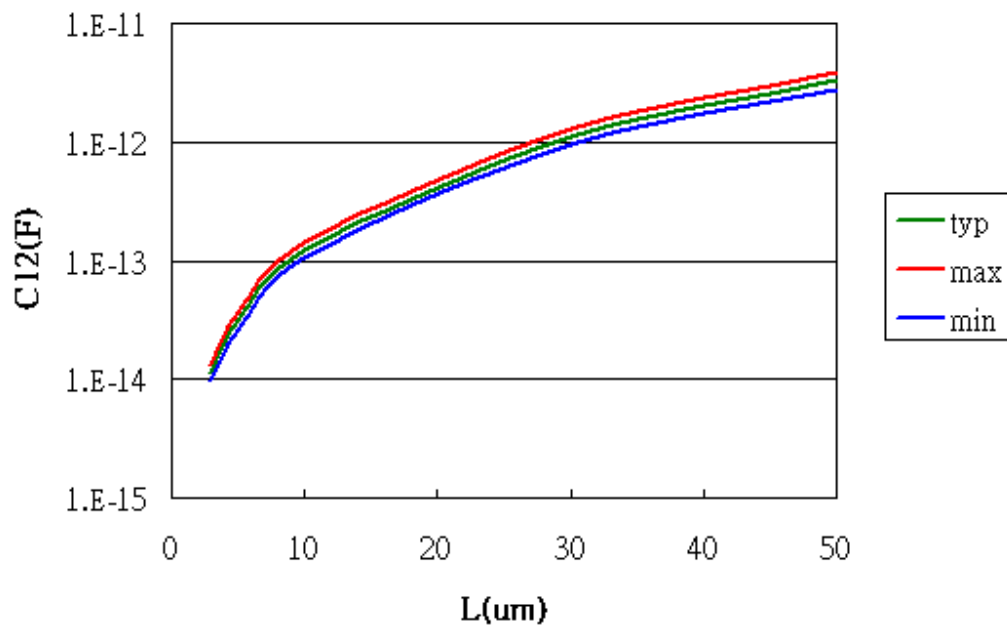


Figure 2-5 Model-predicted main capacitance for 3-corner cases at T=25 °C, frequency=5GHz, W/L ratio is 1, width range is 3um~50um, metal option is 1P7M(mo=21)